

Review Questions

The answers to the chapter review questions can be found in the Appendix.

1. Which code can be inserted to have the code print 2?

```
public class BirdSeed {  
    private int numberBags;  
    boolean call;  
  
    public BirdSeed() {  
        // LINE 1  
        call = false;  
        // LINE 2  
    }  
  
    public BirdSeed(int numberBags) {  
        this.numberBags = numberBags;  
    }  
  
    public static void main(String[] args) {  
        var seed = new BirdSeed();  
        System.out.print(seed.numberBags);  
    } }
```

- A. Replace line 1 with `BirdSeed(2);`.
 - B. Replace line 2 with `BirdSeed(2);`.
 - C. Replace line 1 with `new BirdSeed(2);`.
 - D. Replace line 2 with `new BirdSeed(2);`.
 - E. Replace line 1 with `this(2);`.
 - F. Replace line 2 with `this(2);`.
 - G. The code prints 2 without any changes.
2. Which modifier pairs can be used together in a method declaration? (Choose all that apply.)
- A. `static` and `final`
 - B. `private` and `static`
 - C. `static` and `abstract`
 - D. `private` and `abstract`
 - E. `abstract` and `final`
 - F. `private` and `final`

3. Which of the following statements about methods are true? (Choose all that apply.)

- A. Overloaded methods must have the same signature.
- B. Overridden methods must have the same signature.
- C. Hidden methods must have the same signature.
- D. Overloaded methods must have the same return type.
- E. Overridden methods must have the same return type.
- F. Hidden methods must have the same return type.

4. What is the output of the following program?

```
1: class Mammal {  
2:     private void sneeze() {}  
3:     public Mammal(int age) {  
4:         System.out.print("Mammal");  
5:     } }  
6: public class Platypus extends Mammal {  
7:     int sneeze() { return 1; }  
8:     public Platypus() {  
9:         System.out.print("Platypus");  
10:    }  
11:    public static void main(String[] args) {  
12:        new Mammal(5);  
13:    } }
```

- A. Platypus
- B. Mammal
- C. PlatypusMammal
- D. MammalPlatypus
- E. The code will compile if line 7 is changed.
- F. The code will compile if line 9 is changed.

5. Which of the following complete the constructor so that this code prints out 50? (Choose all that apply.)

```
class Speedster {  
    int numSpots;  
}  
public class Cheetah extends Speedster {  
    int numSpots;  
  
    public Cheetah(int numSpots) {  
        // INSERT CODE HERE  
    }  
}
```

```
public static void main(String[] args) {  
    Speedster s = new Cheetah(50);  
    System.out.print(s.numSpots);  
}  
}
```

- A. numSpots = numSpots;
 - B. numSpots = this.numSpots;
 - C. this.numSpots = numSpots;
 - D. numSpots = super.numSpots;
 - E. super.numSpots = numSpots;
 - F. The code does not compile regardless of the code inserted into the constructor.
 - G. None of the above
6. Which of the following declare immutable classes? (Choose all that apply.)

```
public final class Moose {  
    private final int antlers;  
}
```

```
public class Caribou {  
    private int antlers = 10;  
}
```

```
public class Reindeer {  
    private final int antlers = 5;  
}
```

```
public final class Elk {}
```

```
public final class Deer {  
    private final Object o = new Object();  
}
```

- A. Moose
- B. Caribou
- C. Reindeer
- D. Elk
- E. Deer
- F. None of the above

7. What is the output of the following code?

```
1: class Arthropod {
2:     protected void printName(long input) {
3:         System.out.print("Arthropod");
4:     }
5:     void printName(int input) {
6:         System.out.print("Spooky");
7:     } }
8: public class Spider extends Arthropod {
9:     protected void printName(int input) {
10:        System.out.print("Spider");
11:    }
12:    public static void main(String[] args) {
13:        Arthropod a = new Spider();
14:        a.printName((short)4);
15:        a.printName(4);
16:        a.printName(5L);
17:    } }
```

- A. SpiderSpiderArthropod
- B. SpiderSpiderSpider
- C. SpiderSpookyArthropod
- D. SpookySpiderArthropod
- E. The code will not compile because of line 5.
- F. The code will not compile because of line 9.
- G. None of the above

8. What is the result of the following code?

```
1: abstract class Bird {
2:     private final void fly() { System.out.println("Bird"); }
3:     protected Bird() { System.out.print("Wow-"); }
4: }
5: public class Pelican extends Bird {
6:     public Pelican() { System.out.print("Oh-"); }
7:     protected void fly() { System.out.println("Pelican"); }
8:     public static void main(String[] args) {
9:         var chirp = new Pelican();
10:        chirp.fly();
11:    } }
```

- A. Oh-Bird
 - B. Oh-Pelican
 - C. Wow-Oh-Bird
 - D. Wow-Oh-Pelican**
 - E. The code contains a compilation error.
 - F. None of the above
9. Which of the following statements about overridden methods are true? (Choose all that apply.)
- A. An overridden method must contain method parameters that are the same or covariant with the method parameters in the inherited method.
 - B. An overridden method may declare a new exception, provided it is not checked.**
 - C. An overridden method must be more accessible than the method in the parent class.
 - D. An overridden method may declare a broader checked exception than the method in the parent class.
 - E. If an inherited method returns void, then the overridden version of the method must return void.**
 - F. None of the above
10. Which of the following pairs, when inserted into the blanks, allow the code to compile? (Choose all that apply.)
- ```

1: public class Howler {
2: public Howler(long shadow) {
3: _____;
4: }
5: private Howler(int moon) {
6: super();
7: }
8: }
9: class Wolf extends Howler {
10: protected Wolf(String stars) {
11: super(2L);
12: }
13: public Wolf() {
14: _____;
15: }
16: }
```
- A. this(3) at line 3, this("") at line 14**
  - B. this() at line 3, super(1) at line 14
  - C. this((short)1) at line 3, this(null) at line 14**
  - D. super() at line 3, super() at line 14

- E. `this(2L)` at line 3, `super((short)2)` at line 14
- F. `this(5)` at line 3, `super(null)` at line 14
- G. Remove lines 3 and 14.

11. What is the result of the following?

```
1: public class PolarBear {
2: StringBuilder value = new StringBuilder("t");
3: { value.append("a"); }
4: { value.append("c"); }
5: private PolarBear() {
6: value.append("b");
7: }
8: public PolarBear(String s) {
9: this();
10: value.append(s);
11: }
12: public PolarBear(CharSequence p) {
13: value.append(p);
14: }
15: public static void main(String[] args) {
16: Object bear = new PolarBear();
17: bear = new PolarBear("f");
18: System.out.println(((PolarBear)bear).value);
19: } }
```

- A. `tacb`
- B. `tacf`
- C. `tacbf`
- D. `tcafb`
- E. `taftacb`
- F. The code does not compile.
- G. An exception is thrown.

12. How many lines of the following program contain a compilation error?

```
1: public class Rodent {
2: public Rodent(Integer x) {}
3: protected static Integer chew() throws Exception {
4: System.out.println("Rodent is chewing");
5: return 1;
6: }
```

```
7: }
8: class Beaver extends Rodent {
9: public Number chew() throws RuntimeException {
10: System.out.println("Beaver is chewing on wood");
11: return 2;
12: } }
```

**A.** None

**B.** 1

**C.** 2

**D.** 3

**E.** 4

**F.** 5

- 13.** Which of these classes compile and will include a default constructor created by the compiler? (Choose all that apply.)

**A.**

```
public class Bird {}
```

**B.**

```
public class Bird {
 public bird() {}
}
```

**C.**

```
public class Bird {
 public bird(String name) {}
}
```

**D.**

```
public class Bird {
 public Bird() {}
}
```

**E.**

```
public class Bird {
 Bird(String name) {}
}
```

**F.**

```
public class Bird {
 private Bird(int age) {}
}
```

**G.**

```
public class Bird {
 public Bird bird() { return null; }
}
```

14. Which of the following statements about inheritance are correct? (Choose all that apply.)

- A. A class can directly extend any number of classes.
- B. A class can implement any number of interfaces.
- C. All variables inherit `java.lang.Object`.
- D. If class A is extended by B, then B is a superclass of A.
- E. If class C implements interface D, then C is a subtype of D.
- F. Multiple inheritance is the property of a class to have multiple direct superclasses.

15. Which statements about the following program are correct? (Choose all that apply.)

```
1: abstract class Nocturnal {
2: boolean isBlind();
3: }
4: public class Owl extends Nocturnal {
5: public boolean isBlind() { return false; }
6: public static void main(String[] args) {
7: var nocturnal = (Nocturnal)new Owl();
8: System.out.println(nocturnal.isBlind());
9: } }
```

- A. It compiles and prints true.
- B. It compiles and prints false.
- C. The code will not compile because of line 2.
- D. The code will not compile because of line 5.
- E. The code will not compile because of line 7.
- F. The code will not compile because of line 8.
- G. None of the above

16. What is the result of the following?

```
1: class Arachnid {
2: static StringBuilder sb = new StringBuilder();
3: { sb.append("c"); }
4: static
5: { sb.append("u"); }
6: { sb.append("r"); }
7: }
```

```

8: public class Scorpion extends Arachnid {
9: static
10: { sb.append("q"); }
11: { sb.append("m"); }
12: public static void main(String[] args) {
13: System.out.print(Scorpion.sb + " ");
14: System.out.print(Scorpion.sb + " ");
15: new Arachnid();
16: new Scorpion();
17: System.out.print(Scorpion.sb);
18: } }

```

- A. qu qu qumrcrc
  - B. u u ucrcrm
  - C. uq uq uqmcrcr
  - D. uq uq uqcrcrm
  - E. qu qu qumrcrc
  - F. qu qu qucrcrm**
  - G. The code does not compile.
17. Which of the following are true? (Choose all that apply.)
- A. `this()` can be called from anywhere in a constructor.
  - B. `this()` can be called from anywhere in an instance method.
  - C. `this.variableName` can be called from any instance method in the class.
  - D. `this.variableName` can be called from any static method in the class.
  - E. You can call the default constructor written by the compiler using `this()`.
  - F. You can access a private constructor with the `main()` method in the same class.
18. Which statements about the following classes are correct? (Choose all that apply.)
- ```

1: public class Mammal {
2:     private void eat() {}
3:     protected static void drink() {}
4:     public Integer dance(String p) { return null; }
5: }
6: class Primate extends Mammal {
7:     public void eat(String p) {}
8: }
9: class Monkey extends Primate {

```

```
10: public static void drink() throws RuntimeException {}
11: public Number dance(CharSequence p) { return null; }
12: public int eat(String p) {}
13: }
```

- A. The eat() method in Mammal is correctly overridden on line 7.
- B. The eat() method in Mammal is correctly overloaded on line 7.
- C. The drink() method in Mammal is correctly overridden on line 10.
- D. The drink() method in Mammal is correctly hidden on line 10.
- E. The dance() method in Mammal is correctly overridden on line 11.
- F. The dance() method in Mammal is correctly overloaded on line 11.
- G. The eat() method in Primate is correctly hidden on line 12.
- H. The eat() method in Primate is correctly overloaded on line 12.

19. What is the output of the following code?

```
1: class Reptile {
2:     {System.out.print("A");}
3:     public Reptile(int hatch) {}
4:     void layEggs() {
5:         System.out.print("Reptile");
6:     } }
7: public class Lizard extends Reptile {
8:     static {System.out.print("B");}
9:     public Lizard(int hatch) {}
10:    public final void layEggs() {
11:        System.out.print("Lizard");
12:    }
13:    public static void main(String[] args) {
14:        var reptile = new Lizard(1);
15:        reptile.layEggs();
16:    } }
```

- A. AALizard
- B. BALizard
- C. BLizardA
- D. ALizard
- E. The code will not compile because of line 3.
- F. None of the above

20. Which statement about the following program is correct?

```
1: class Bird {  
2:     int feathers = 0;  
3:     Bird(int x) { this.feathers = x; }  
4:     Bird fly() {  
5:         return new Bird(1);  
6:     } }  
7: class Parrot extends Bird {  
8:     protected Parrot(int y) { super(y); }  
9:     protected Parrot fly() {  
10:         return new Parrot(2);  
11:     } }  
12: public class Macaw extends Parrot {  
13:     public Macaw(int z) { super(z); }  
14:     public Macaw fly() {  
15:         return new Macaw(3);  
16:     }  
17:     public static void main(String... sing) {  
18:         Bird p = new Macaw(4);  
19:         System.out.print(((Parrot)p.fly()).feathers);  
20:     } }
```

- A. One line contains a compiler error.
 - B. Two lines contain compiler errors.
 - C. Three lines contain compiler errors.
 - D. The code compiles but throws a `ClassCastException` at runtime.
 - E. The program compiles and prints 3.
 - F. The program compiles and prints 0.
21. Which of the following are properties of immutable classes? (Choose all that apply.)
- A. The class can contain setter methods, provided they are marked `final`.
 - B. The class must not be able to be extended outside the class declaration.
 - C. The class may not contain any instance variables.
 - D. The class must be marked `static`.
 - E. The class may not contain any `static` variables.
 - F. The class may only contain `private` constructors.
 - G. The data for mutable instance variables may be read, provided they cannot be modified by the caller.

22. What does the following program print?

```
1: class Person {
2:     static String name;
3:     void setName(String q) { name = q; } }
4: public class Child extends Person {
5:     static String name;
6:     void setName(String w) { name = w; }
7:     public static void main(String[] p) {
8:         final Child m = new Child();
9:         final Person t = m;
10:        m.name = "Elysia";
11:        t.name = "Sophia";
12:        m.setName("Webby");
13:        t.setName("Olivia");
14:        System.out.println(m.name + " " + t.name);
15:    } }
```

- A. Elysia Sophia
- B. Webby Olivia
- C. Olivia Olivia**
- D. Olivia Sophia
- E. The code does not compile.
- F. None of the above

23. What is the output of the following program?

```
1: class Canine {
2:     public Canine(boolean t) { logger.append("a"); }
3:     public Canine() { logger.append("q"); }
4:
5:     private StringBuilder logger = new StringBuilder();
6:     protected void print(String v) { logger.append(v); }
7:     protected String view() { return logger.toString(); }
8: }
9:
10: class Fox extends Canine {
11:     public Fox(long x) { print("p"); }
12:     public Fox(String name) {
13:         this(2);
14:         print("z");
15:     }
```

```
16: }
17:
18: public class Fennec extends Fox {
19:     public Fennec(int e) {
20:         super("tails");
21:         print("j");
22:     }
23:     public Fennec(short f) {
24:         super("eevee");
25:         print("m");
26:     }
27:
28:     public static void main(String... unused) {
29:         System.out.println(new Fennec(1).view());
30:     } }
```

- A. qpz
- B. qpzj
- C. jzpa
- D. apj**
- E. apjm
- F. The code does not compile.
- G. None of the above

24. What is printed by the following program?

```
1: class Antelope {
2:     public Antelope(int p) {
3:         System.out.print("4");
4:     }
5:     { System.out.print("2"); }
6:     static { System.out.print("1"); }
7: }
8: public class Gazelle extends Antelope {
9:     public Gazelle(int p) {
10:         super(6);
11:         System.out.print("3");
12:     }
```

```
13: public static void main(String hopping[]) {  
14:     new Gazelle(0);  
15: }  
16: static { System.out.print("8"); }  
17: { System.out.print("9"); }  
18: }
```

- A. 182640
- B. 182943
- C. 182493
- D. 421389
- E. The code does not compile.
- F. The output cannot be determined until runtime.

25. Which of the following are true about a concrete class? (Choose all that apply.)

- A. A concrete class can be declared as `abstract`.
- B. A concrete class must implement all inherited abstract methods.
- C. A concrete class can be marked as `final`.
- D. A concrete class must be immutable.
- E. A concrete method that implements an abstract method must match the method declaration of the abstract method exactly.

26. What is the output of the following code?

```
4: public abstract class Whale {  
5:     public abstract void dive();  
6:     public static void main(String[] args) {  
7:         Whale whale = new Orca();  
8:         whale.dive(3);  
9:     }  
10: }  
11: class Orca extends Whale {  
12:     static public int MAX = 3;  
13:     public void dive() {  
14:         System.out.println("Orca diving");  
15:     }  
16:     public void dive(int... depth) {  
17:         System.out.println("Orca diving deeper " + MAX);  
18:     } }
```

- A. Orca diving
- B. Orca diving deeper 3
- C. The code will not compile because of line 4.
- D. The code will not compile because of line 8.
- E. The code will not compile because of line 11.
- F. The code will not compile because of line 12.
- G. The code will not compile because of line 17.
- H. None of the above